

$k = 3$ Alfeld sharpness on general smooth surfaces: indefinite and parabolic curvature via the edge-normal moves

subagent report

notes/v6/td.k3b, 2026-09-28

Summary and status

Statement	Status
Per-face moment formula on every edge star, any geometry and weights: the moment of a Γ -face is an explicit linear function of the vertex jet δ and the spoke data $P_x = \delta + p_x$, modulo the single-macro line $\mathbb{R}D_F$ (Lemma 1.1)	[PROVED] (exact reference computation + transport); 29/29 faces agree exactly with the reduced model
Flat reduction: in the Γ -flat limit the admissible parameters are δ , one free normal datum ν_j per spoke, and a 3-parameter flux datum $\sigma_j = \gamma - s_j \wedge \eta$ (Lemma 2.1)	[PROVED]; identities checked symbolically
Star Nondegeneracy Theorem: on a Γ -flat edge star with a common quadratic form $\mathfrak{h} \neq 0$, the tangential total moment has rank 2 unless <i>every</i> Γ -face of the star is degenerate at z . A face is degenerate at z iff $\mathfrak{h} = \pm \nu \otimes \nu$ has rank one and $\nu \perp$ the edge opposite z (Thm. 3.3, Lemma 3.2)	[PROVED]; exhaustive-in- $\mathfrak{h}$ exact checks on 3034 stars, no exception
Every Γ -face is nondegenerate at the vertex opposite its edge of largest $ \mathfrak{h}(e, e) $. Uniform witness $t \cdot \sum \mathcal{M} \geq \kappa_0 \mathfrak{h} (\text{diam } F)^{5/2} - C(\text{diam } F)^{7/2}$ for all $t \in T\Gamma$ (Thm. 4.2)	[PROVED]; $\kappa_0(\theta) > 0$ by compactness (not explicit)
Main result: for $k \geq 3$ on Alfeld refinements, $\ \nabla(\tilde{u} - u_h)\ \geq c h_\Gamma^{3/2} \ \mathfrak{h}\ \partial_N u \ _{L^2(\Gamma)} - C h_\Gamma^2$. This is Thm. B:lb, now with $k \geq 3$ in place of $k \geq 4$ (Thm. 4.3)	[PROVED]. Assumptions as Thm. B:lb; the upper bound keeps its td:open item 3 caveat
Referee 8 MINOR FIX (the $t \parallel F$ step): the patch lemma is restated with t tangent to Γ (Lemma 4.1)	[PROVED]
Full-FE cross-check (patchlib): rank predictions confirmed on 7 stars, including $\sum c_i = 0$ with rank 2 and a degenerate pair with rank 1	[NUMERICAL] (mod- p ranks)

Main finding. The vertex move of td.k3 sees only $\sum_i c_i \propto \sum_i |F_i| \mathfrak{h}(e_{ab}, e_{ab})/H_i$, and this sum can vanish for indefinite $\mathfrak{h}$. The Ring Trace Theorem has two further families of moves:

- the *normal spoke data* ν_j , one free scalar per spoke;
- the *flux data* σ_j , an affine function of the spoke vector.

Their moments are weighted by the *differences* $\mathfrak{h}(s_a, s_a) - \mathfrak{h}(s_b, s_b)$ of the two spoke weights of a face. These differences are nonzero exactly where the vertex move loses information.

Combined with the single-macro directions D_F , the image of the moment map is degenerate only at *parabolic* points: $\mathfrak{h} = \pm\nu \otimes \nu$, with an edge of F aligned with the asymptotic direction $\nu^\perp$. At a saddle point ($\det \mathfrak{h} < 0$) and at an elliptic point every edge star is nondegenerate. At a parabolic point, the edge star at a different vertex of the same face is nondegenerate.

The earlier `td_k3` remark was too optimistic about the definite-curvature case. The single edge star at a chosen vertex can be degenerate for the *semidefinite* form $\mathfrak{h} = (x + y)^2$ (`degen.log`, “flat square, 1”). The vertex choice of Thm. 4.2 is therefore needed.

Conventions: as in `td_k3/td_sharp` Part B. $\Gamma \in C^3$, and Γ_h is inscribed with (G1)–(G3). $\mathcal{T}_h$ is the Alfeld refinement of a shape-regular macro mesh (minimal angle θ). $W = \partial_N u|_\Gamma$, and $\mathfrak{h}_y$ is the second fundamental form. The sagitta weights are $w_{ij}^F = \mathfrak{h}_F(e_{ij}, e_{ij})$, up to $O(\ell^3)$ (Lemma B:face(a)). $\mathcal{M}_F(v) = \frac{4}{H} \int_F Q_F \hat{g} dA$ with $Q_F = \frac{1}{2} \sum w_{ij} \mu_i \mu_j$, as in `td_k3` Sec. 2.

1 The per-face moment formula

Let $R = R(z, q)$ be an edge star with link $x_0, \dots, x_{m-1}$, Γ -faces $F_i = \text{conv}(z, x_i, x_{i+1})$ for $i \in I$, and data (δ, p_q, p_j, Φ) as in Lemma l:nec. For a face $F = F_i = \text{conv}(z, a, b)$ put $e_a = a - z$ and $e_b = b - z$. Put $P_a = \delta + p_a$ and $P_b = \delta + p_b$, the spoke data of the two spokes of F . Let $u_a = \nabla \lambda_q^{s_3}$ and $u_b = \nabla \lambda_q^{s_2}$ ($s_3 = \text{conv}(B, q, z, a)$, $s_2 = \text{conv}(B, q, z, b)$). Write $w = (w_{za}, w_{zb}, w_{ab})$ for the weights of F , and

$$D_F := (w_{za} - w_{ab}) e_a + (w_{ab} - w_{zb}) e_b \quad (\text{the single-macro direction, Prop. A:k3}).$$

Lemma 1.1 (Per-face moment). *For every $v \in Y^1(R)$ with interior-face data (δ, p_q, p_j, Φ) and every $i \in I$,*

$$\mathcal{M}_{F_i}(v) \equiv \frac{8|F_i|}{H_i} \left[-\frac{w_{ab}}{2880} G_i \delta - \frac{w_{za} - w_{zb}}{2880} \left((P_a \cdot u_a) e_b - (P_b \cdot u_b) e_a \right) \right] \pmod{\mathbb{R} D_{F_i}}.$$

Conversely, every value of the right-hand side, plus any multiple of D_{F_i} chosen independently for each i , is attained by some $v \in Y^1(R)$ with these data.

Proof. Reference computation (`ref_nu.py`, exact sympy). Take the reference triangle $z = 0$, $a = e_1$, $b = e_2$. For $\hat{g} \in U_F \cap \{\int \hat{g} = 0\}$ with $\hat{g}(z) = e$, $\tilde{\alpha} := \hat{g}(m_{za}) \cdot \nabla \mu_b$ and $\tilde{\beta} := \hat{g}(m_{zb}) \cdot \nabla \mu_a$, the moment is determined modulo the line D of Lemma l:K. For one particular representative, linear in $(e, \tilde{\alpha}, \tilde{\beta})$, the $\tilde{\alpha}$ - and $\tilde{\beta}$ -columns of $\int Q \hat{g}$ are $\frac{w_{ab} - w_{za}}{180} (\hat{e}_1 - \hat{e}_2)$ and $\frac{w_{zb} - w_{za}}{180} \hat{e}_1$. The e -columns are those of K in Lemma l:K. Adding $4D$ (with $720D = (w_{za} - w_{ab}) \hat{e}_1 + (w_{ab} - w_{zb}) \hat{e}_2$) shows $\frac{w_{ab} - w_{za}}{180} (\hat{e}_1 - \hat{e}_2) \equiv \frac{w_{za} - w_{zb}}{180} \hat{e}_2$ modulo D . Together with Lemma l:K this gives

$$\int_{\text{ref}} Q \hat{g} \equiv -\frac{w_{ab}}{720} e + \frac{w_{za} - w_{zb}}{180} (\tilde{\alpha} \hat{e}_2 - \tilde{\beta} \hat{e}_1) \pmod{D}.$$

Transport. Write $\hat{g} = A \hat{g}_{\text{ref}}$ with $A \hat{e}_1 = e_a$ and $A \hat{e}_2 = e_b$. Then $\tilde{\alpha}, \tilde{\beta}$ are affine invariants ($\hat{g}_{\text{ref}} \cdot \hat{e}_2 = \hat{g} \cdot A^{-\top} \hat{e}_2 = \hat{g} \cdot \nabla \mu_b$). So

$$\int_F Q \hat{g} \equiv 2|F| \left[-\frac{w_{ab}}{720} \hat{g}(z) + \frac{w_{za} - w_{zb}}{180} (\tilde{\alpha} e_b - \tilde{\beta} e_a) \right] \pmod{\mathbb{R} D_F},$$

and $AD_{\text{ref}} \propto D_F$.

Traces. $\hat{g}(z) = \frac{1}{4} G_i \delta$ (Lemma l:jet). On $[z, a]$, Lemma l:nec(d) reads $\hat{g} \cdot \nabla \lambda_B^{s_3} = -\frac{1}{4} P_a \cdot u_a$ at m_{za} , since $h_i(m_{za}) = \frac{1}{4}(\delta + p_a)$. Here $\lambda_B^{s_3}$ is linear, vanishes on the plane of $\text{conv}(q, z, a)$, and equals 1 at $B = \frac{1}{4}(q + z + a + b)$, so it equals 4 at b . Hence the tangential part of $\nabla \lambda_B^{s_3}$ is orthogonal

to e_a and pairs to 4 with e_b , i.e. it equals $4\nabla\mu_b$. So $\tilde{\alpha} = -\frac{1}{16}P_a \cdot u_a$, and likewise $\tilde{\beta} = -\frac{1}{16}P_b \cdot u_b$. Multiply by $\frac{4}{H_i}$.

Converse. Thm. t:trace and Lemma l:suff give realisability. Lemma l:single makes the means zero and leaves exactly the D_{F_i} -freedom on each face independently. $\square$

Checks. `face_formula.py`, exact, reduced model: random data (δ, p_q, p_j, Φ) with equal fluxes, generic weights, both Γ -flat and curved stars with $m = 3, \dots, 6$ and arbitrary I . The prediction is compared *face by face* modulo the single line $\mathbb{R}D_{F_i}$, so the test is never vacuous (this answers the Referee 8 caveat on D-span tests). Result: 29/29 faces agree (`face_formula.log`). The term $-\frac{8|F|w_{ab}}{2880H}G\delta = -\frac{c_i}{360}G_i\delta$ is Thm. t:vertex.

2 The Γ -flat reduction

A star is Γ -flat if its Γ -faces lie in one plane T . Put $z = 0$, $T = \{x_3 = 0\}$ and $q = (q', H)$ with $H > 0$. The non- Γ link vertices are unrestricted. Call a spoke $s_j = x_j$ a Γ -spoke if it bounds a Γ -face; then $s_j \in T$. Let J be the rotation by $+\frac{\pi}{2}$ in T . Orient so that $s_i \wedge s_{i+1} = 2|F_i| > 0$ for $i \in I$; this holds because the macro-tetrahedra around $[z, q]$ are consistently oriented and q lies on one side of T . Put $\omega_j := w(s_j) := \mathfrak{h}(s_j, s_j)$, $w_i := w_{ab}^{F_i}$ and $\hat{\tau}_i := \omega_i - \omega_{i+1} = w_{za} - w_{zb}$. Write $P_j = (\pi_j, \nu_j) \in T \times \mathbb{R}$ and

$$\sigma_j := Js_j \cdot (\pi_j - \nu_j q' / H).$$

Lemma 2.1 (Flat reduction). *On a Γ -flat star with weights $w_{ij}^F = \mathfrak{h}(e_{ij}, e_{ij})$ for one symmetric form $\mathfrak{h}$ on T :*

(a) $P_a \cdot u_a = -\sigma_a/(2|F|) + \nu_a/H$ and $P_b \cdot u_b = \sigma_b/(2|F|) + \nu_b/H$. Hence, modulo $\text{span}\{D_{F_i}\}$,

$$\sum_{i \in I} \mathcal{M}_{F_i} \equiv \frac{1}{360H} \sum_{i \in I} \left[-|F_i| w_i G_i \delta + \hat{\tau}_i \left(\frac{1}{2}(\sigma_i s_{i+1} + \sigma_{i+1} s_i) - \frac{|F_i|}{H}(\nu_i s_{i+1} - \nu_{i+1} s_i) \right) \right].$$

(b) *The flux condition for spoke j is $A_j \cdot P_j = \frac{H}{2}\sigma_j$. The admissible spoke data are exactly: $\nu_j \in \mathbb{R}$ free for each Γ -spoke, and $\sigma_j = \gamma - s_j \wedge \eta$ with $(\gamma, \eta) \in \mathbb{R} \times \mathbb{R}^2$ free. The data $\delta \in \mathbb{R}^3$, (ν_j) , (γ, η) and the D-coefficients are mutually independent.*

Proof. (a) In the flat case $u_a = (-\nabla\mu_b, (1 + \nabla\mu_b \cdot q')/H)$, $u_b = (-\nabla\mu_a, (1 + \nabla\mu_a \cdot q')/H)$, $\nabla\mu_b = Js_a/(2|F|)$ and $\nabla\mu_a = -Js_b/(2|F|)$. Substitute into Lemma 1.1; $H_i = H$ for all i .

(b) $A_j = \frac{1}{2}q \times x_j = \frac{1}{2}(HJs_j, q' \wedge s_j)$ gives $A_j \cdot P_j = \frac{H}{2}\sigma_j$. Lemma l:nec(b) with $p_j = P_j - \delta$ gives $A_j \cdot P_j = 60\Phi - A_j \cdot \psi$ with $\psi := \delta + 2p_q$. Since p_q and Φ are free, so are ψ and Φ . Then $\sigma_j = \frac{120\Phi}{H} - s_j \wedge (\psi' - \psi_n q' / H)$. Given ν_j and σ_j , the component $Js_j \cdot \pi_j$ is fixed ($Js_j \neq 0$); the component of π_j along s_j is invisible. Spokes that bound no Γ -face enter only through their own flux equation. $\square$

Check: `flat_algebra.py` (symbolic in a, b, q, P): (a), the flux identity and the affine form of σ all hold (`flat_algebra.log`).

3 Star Nondegeneracy Theorem

Definition 3.1. A Γ -face $F = \text{conv}(z, a, b)$ of a Γ -flat star is *degenerate at z* (for $\mathfrak{h}$) if $w_{ab} = \mathfrak{h}(b - a, b - a) = 0$ and $\mathfrak{h}(e_a, e_a) = \mathfrak{h}(e_b, e_b)$. This is the rank-1 case of Cor. c:single(c).

Lemma 3.2. *F is degenerate at z iff $\mathfrak{h} = 0$ or $\mathfrak{h} = \pm \nu \otimes \nu$ with $\nu \perp b - a$. In particular F is not degenerate at any vertex if $\det \mathfrak{h} \neq 0$. If $\mathfrak{h} \neq 0$, F is nondegenerate at the vertex opposite any edge e of F with $\mathfrak{h}(e, e) \neq 0$, and such an edge exists.*

Proof. Put $e = b - a$. Then $\mathfrak{h}(e_b) = \mathfrak{h}(e_a) + 2\mathfrak{h}(e_a, e) + \mathfrak{h}(e)$, so the conditions say $\mathfrak{h}(e, e) = \mathfrak{h}(e_a, e) = 0$.

If $\det \mathfrak{h} \neq 0$, the $\mathfrak{h}$ -orthogonal complement of a null vector e is $\mathbb{R}e$. Then $e_a \parallel e$, which is impossible for a triangle.

If $\mathfrak{h} = \pm \nu \otimes \nu$, both conditions reduce to $\nu \cdot e = 0$.

A quadratic form on $\mathbb{R}^2$ vanishing on the three pairwise non-parallel edge vectors of F is 0. $\square$

Theorem 3.3 (Star Nondegeneracy). *Let R be a Γ -flat edge star, $\mathfrak{h} \neq 0$ a symmetric form on T, and weights $w_{ij}^F = \mathfrak{h}(e_{ij}, e_{ij})$. Assume the Γ -spokes are pairwise non-parallel in the same direction; on meshes they are separated by angles $\geq \theta/2$. If some Γ -face of R is nondegenerate at z, then the tangential total moment $v \mapsto \sum_{i \in I} \mathcal{M}_{F_i}(v)$ maps $Y^1(R)$ onto T.*

Proof. Suppose $0 \neq \lambda \in T$ annihilates the image. Put $\ell_j = \lambda \cdot s_j$, let $\mu := J\lambda$, and set $\tau_i := |F_i| \hat{\tau}_i$. By Lemma 2.1 the parameters $\delta, \nu_j, \gamma, \eta$ and the D-coefficients are independent, so λ annihilates each family separately:

$$\begin{aligned} (\delta_T) \quad & \sum_{i \in I} |F_i| w_i = 0 \quad (\delta \in T, G_i|_T = I); & (D) \quad & \lambda \cdot D_{F_i} = 0 \quad \forall i \in I; \\ (\nu_j) \quad & \tau_{j-1} \ell_{j-1} \mathbf{1}_{j-1} = \tau_j \ell_{j+1} \mathbf{1}_j \quad \text{for every } \Gamma\text{-spoke } j \quad (\mathbf{1}_i = [i \in I]); \\ (\sigma) \quad & \sum_{i \in I} \hat{\tau}_i [\gamma(\ell_i + \ell_{i+1}) + L(s_i) \ell_{i+1} + L(s_{i+1}) \ell_i] = 0 \quad \forall \gamma \in \mathbb{R}, L \in T^*. \end{aligned}$$

Step 1: $\tau_i = 0$ for all $i \in I$.

Put $\varpi_i := \tau_i \ell_{i+1}$. Multiplying (ν_j) by ℓ_j gives $\varpi_{j-1} = \varpi_j$ when both faces are in I, and $\varpi_j = 0$ at the first face of a run of consecutive Γ -faces. So ϖ is constant on each run and zero on non-cyclic runs. If I is the whole cycle, (σ) with $\gamma = 0, L = \lambda \cdot$ gives $2 \sum \varpi_i / |F_i| = 0$, so again $\varpi = 0$. Hence $\tau_i = 0$ whenever $\ell_i \ell_{i+1} \neq 0$.

Let Z be the set of Γ -spokes with $\ell_p = 0$, i.e. $s_p \parallel \mu$. By the separation assumption, $|Z| \leq 2$. If $|Z| = 2$, the two spokes point in opposite directions. No face has both spokes in Z.

For $p \in Z$, (ν_p) defines $X_p := \tau_{p-1} \ell_{p-1} \mathbf{1}_{p-1} = \tau_p \ell_{p+1} \mathbf{1}_p$. If one of F_{p-1}, F_p is not in I, then $X_p = 0$. The only possibly nonzero τ 's are $\tau_{p-1} = X_p / \ell_{p-1}$ and $\tau_p = X_p / \ell_{p+1}$.

Now (σ) with $L = 0$ and with $L = \mu \cdot$ gives, with $k_p := \frac{1}{|F_{p-1}|} + \frac{1}{|F_p|} > 0$:

$$\sum_{p \in Z} k_p X_p = 0, \quad \sum_{p \in Z} (\mu \cdot s_p) k_p X_p = 0.$$

The numbers $\mu \cdot s_p$ ($p \in Z$) are nonzero and, if there are two, of opposite signs. So $X_p = 0$ for all $p \in Z$, and $\tau \equiv 0$.

Step 2: $w_i = 0$ for all $i \in I$.

Now $\omega_i = \omega_{i+1} =: \Omega_i$, and (D) reads $(\Omega_i - w_i)(\ell_i - \ell_{i+1}) = 0$. Suppose $w_i \neq 0$ for some i . There are two cases.

- If $w_i = \Omega_i$, then $\mathfrak{h}$ takes the same nonzero value c on s_i, s_{i+1} and $s_{i+1} - s_i$. In the basis (s_i, s_{i+1}) , $\mathfrak{h} = c \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$, which is definite. Then all $w_{i'}$ ($i' \in I$) are nonzero with the sign of c , contradicting (δ_T) .
- Otherwise $\ell_i = \ell_{i+1}$, i.e. $s_{i+1} - s_i \parallel \mu$, and $w_i = |s_{i+1} - s_i|^2 \mathfrak{h}(\hat{\mu}, \hat{\mu})$.

So every i with $w_i \neq 0$ is of the second kind, and all such w_i have the sign of $\mathfrak{h}(\hat{\mu}, \hat{\mu})$. Then (δ_T) forces them all to vanish.

So every Γ -face is degenerate at z , contradicting the hypothesis. $\square$

Remark 3.4. (i) The proof uses only four families: the vertex moves $\delta \in T$ (via $\sum |F_i| w_i$); the normal spoke moves ν_j ; the flux moves (γ, η) ; and the single-macro directions D_F . It does not use the normal vertex move $(\delta \parallel n)$.

(ii) Converse: if every Γ -face is degenerate at z (so $\mathfrak{h} = \pm \nu \otimes \nu$ with all opposite edges $\perp \nu$), the rank is 1. This is seen in the exact search, e.g. `degen.log` “flat square, 2 opp”, and in the FE check; it is not needed.

(iii) The theorem explains the sphere result of `td_k3`: there $\mathfrak{h} = -I$ and every face is nondegenerate at every vertex. It also locates the only obstruction: parabolic points with an edge along the asymptotic direction.

Checks (exact, exhaustive in $\mathfrak{h}$). `degen.py` computes, for a given star, *all* real $\mathfrak{h} \neq 0$ for which the tangential total moment on the full reduced model has rank < 2 . It writes the moment matrix as $h_{11}M_1 + h_{12}M_2 + h_{22}M_3$, eliminates $\mathfrak{h}$ through 3×3 minors, and takes the real roots in λ .

The theorem’s prediction was compared on:

- 7 structured stars (`degen.log`);
- 1783 random fully flat stars on a $\frac{1}{3}$ -lattice with all subsets I and $|I| = 1..6$ (`degen_random.log`);
- 1251 random Γ -flat stars whose non- Γ link vertices lie off the plane (`degen_gflat.log`).

Every degeneracy found has all Γ -faces individually degenerate, including adjacent pairs such as $I = [1, 2]$ with $\mathfrak{h} = (x + y)^2$. There is no counterexample.

FE cross-check (fe_cross.log, full Alfeld FE, mod- p ranks):

- square fan with all 4 faces and the saddle $\mathfrak{h} = 2xy$, where $\sum_i w_{ab} = 0$ and the vertex move is blind: rank 2;
- single face with $\mathfrak{h} = x^2 - y^2$ ($\sum w_{ab} = 0$): rank 2;
- faces 0, 2 with $\mathfrak{h} = (x + y)^2$ (both degenerate): rank 1;
- faces 0, 1 with the same $\mathfrak{h}$ (face 1 nondegenerate): rank 2;
- a Γ -flat star with off-plane vertices: rank 2.

4 Uniform witness and the lower bound

Lemma 4.1 (Patch form, t tangent to Γ ; Referee 8 fix). *Suppose there are $\kappa_0, K_{\text{ov}}, C_s, C_1$ with the following property. For every $F \in \mathcal{F}_h^\Gamma$ and unit $t \in T_{y_F}\Gamma$ ($y_F = \pi(x_F)$) there is $v_{F,t} \in Y_h^1$ with:*

- $\|\nabla v_{F,t}\| = 1$;
- support in a union ω_F of tetrahedra within distance $C_s h_\Gamma$ of F , with each tetrahedron in at most K_{ov} of the ω_F ;
- $t \cdot \sum_{F' \subset \omega_F \cap \Gamma_h} \mathcal{M}_{F'}(v_{F,t}) \geq \kappa_0 |\mathfrak{h}_{y_F}| (\text{diam } F)^{5/2} - C_1 (\text{diam } F)^{7/2}$.

Then, under the assumptions of Thm. B:main, if $\mathfrak{h}W \neq 0$ and $h_\Gamma \leq h_1$ (h_1 depending also on $\|W\|_{W^{1,\infty}}$ and $\|\mathfrak{h}|W\|_{L^2}$),

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \|\mathfrak{h}|W\|_{L^2(\Gamma)} - C h_\Gamma^2.$$

Proof. Let $W_F := W(y_F) \in T_{y_F}\Gamma$ and $t_F = W_F/|W_F|$. Set $v_F := |\mathfrak{h}_{y_F}| |W_F| v_{F,t_F}$ and $v = \sum_F v_F$. Since t_F is already tangent, no projection onto the plane of F is needed; this is the fix. By Lemma A:lead with B:face(a),

$$\ell_W(v) = \sum_F \left[W_F \cdot \sum_{F'} \mathcal{M}_{F'}(v_F) + \sum_{F'} (W_{F'} - W_F) \cdot \mathcal{M}_{F'}(v_F) \right].$$

We have $|\mathcal{M}_{F'}(v)| \leq C h_\Gamma^{5/2} \|\nabla v\|$ and $|W_{F'} - W_F| \leq C C_s h_\Gamma \|W\|_{W^{1,\infty}}$. So

$$\ell_W(v) \geq \kappa_0 c_0^{5/2} h_\Gamma^{5/2} \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2 - C h_\Gamma^{7/2} \sum_F |\mathfrak{h}_{y_F}| |W_F| (|W_F| + \|W\|_{W^{1,\infty}}).$$

Also $\|\nabla v\|^2 \leq K_{\text{ov}} \sum_F |\mathfrak{h}_{y_F}|^2 |W_F|^2$. By the Riemann sums of Lemma A:assemble (with $|\mathfrak{h}|$ Lipschitz, $\Gamma \in C^3$, Jacobian $1 + O(h_\Gamma^2)$ from B:face(c)):

- $\sum_F |F| |\mathfrak{h}_{y_F}|^2 |W_F|^2 = \|\mathfrak{h}|W\|^2 + O(h_\Gamma)$;
- the error sum is $\leq C h_\Gamma^{-2}$.

Hence $\ell_W(v)/\|\nabla v\| \geq c h_\Gamma^{3/2} \|\mathfrak{h}|W\| - C h_\Gamma^{5/2} / \|\mathfrak{h}|W\|$, and the second term is at most half the first for $h_\Gamma \leq h_1$. Conclude with Cor. A:abstract and Lemma A:lead, whose remainders give the $-C h_\Gamma^2$. $\square$

For $F \in \mathcal{F}_h^\Gamma$ let z_F be the vertex of F opposite an edge e_F that maximises $|\mathfrak{h}_{y_F}(Pe, Pe)|$ over the three edges. Put $\omega_F := R(z_F, q_F)$.

Theorem 4.2 (Uniform witness, $k = 3$ Alfeld, general surfaces). *There are $\kappa_0 = \kappa_0(\theta, \theta_0) > 0$, $C_1 = C_1(\theta, \theta_0, \Gamma)$ and h_* such that for $h_\Gamma \leq h_*$ the hypothesis of Lemma 4.1 holds with $\omega_F = R(z_F, q_F)$, $K_{\text{ov}}, C_s \leq C(\theta)$.*

Proof. Put $\ell = \text{diam } F$ and $\mathfrak{h}_0 := \mathfrak{h}_{y_F} \circ P_{y_F}$.

(1) *Weights.* Every Γ -face F' of ω_F lies within $C\ell$ of y_F . So $w_{ij}^{F'} = \mathfrak{h}_0(e_{ij}, e_{ij}) + O(\ell^3)$ by B:face(a), using $\Gamma \in C^3$ and $|P_y - P_{y_F}| \leq C\ell$. The moments are linear in the weights and $|\mathcal{M}_{F'}(v)| \leq C|w|\ell^{1/2}\|\nabla v\|$ (td_sharp, proof of A:assemble). So replacing the weights by $\mathfrak{h}_0(e, e)$ costs $\leq C\ell^{7/2}\|\nabla v\|$. This is the C_1 term. If $\mathfrak{h}_0 = 0$ there is nothing to prove. Otherwise write $\mathfrak{h}_0 = |\mathfrak{h}_0| \hat{\mathfrak{h}}$ with $|\hat{\mathfrak{h}}| = 1$.

(2) *Parameters and energy.* For a parameter vector $x = (\delta, p_q, \Phi, (p_j), (s_i))$ satisfying the flux equations, Lemma l:suff plus the single-macro adjustments give $v(x) \in Y^1(R)$. By the proof of Thm. t:vertex (fixed reference right inverses, transported affinely), $\|\nabla v(x)\| \leq C_E(\theta)|x|\ell^{1/2}$ in the scaling where x carries units of $|\nabla v|$. Let $\Pi(\mathcal{C}; \hat{\mathfrak{h}})$ be the linear map $x \mapsto t$ -components of $\ell^{-3} \sum \mathcal{M}_{F'}(v(x))$ for weights $\hat{\mathfrak{h}}(e, e)$. By Thm. t:trace and Lemma 1.1, Π is an explicit rational function of the normalised geometry $\mathcal{C}$ (star scaled by $1/\ell$), linear in $\hat{\mathfrak{h}}$. It is defined on the admissible subspace, which has constant dimension because the flux equations have full rank m . So $s(\mathcal{C}, \hat{\mathfrak{h}}) := \sigma_{\min}(\Pi)$ is continuous.

(3) *Compactness.* Let $\mathcal{K}$ be the set of normalised Γ -flat configurations with the following properties:

- $m \leq m_{\max}(\theta)$ and $I \ni i_0$, where $F = F_{i_0}$ has diameter 1;
- all macro-tetrahedra have minimal angle $\geq \theta/2$;
- the Γ -spokes are separated by angles $\geq \theta_0/2$ and consistently oriented;
- $|\hat{\mathfrak{h}}| = 1$ and $|\hat{\mathfrak{h}}(\hat{e}_F, \hat{e}_F)| \geq c_1(\theta_0)$.

$\mathcal{K}$ is compact, since there are finitely many combinatorial types and all conditions are closed. On $\mathcal{K}$ the face F_{i_0} is nondegenerate at z (Lemma 3.2), so $s > 0$ by Thm. 3.3. By compactness, $s \geq s_0(\theta, \theta_0) > 0$.

The constant c_1 exists because $\mathfrak{h} \mapsto (\mathfrak{h}(e, e))_{e \in \text{edges}}$ is invertible with inverse bounded by $C(\theta_0)\ell^{-2}$ (Lemma B:face(d)). So the maximising edge has $|\hat{\mathfrak{h}}(\hat{e}_F, \hat{e}_F)| \geq c_1$.

(4) *Actual meshes.* Project the Γ -vertices of ω_F onto $T_{y_F}\Gamma$. This moves them by $O(\ell^2)$, i.e. $O(\ell)$ after normalisation, so the actual normalised configuration is within $O(\ell)$ of a point of $\mathcal{K}$. The conditions hold in the limit by (G1), (G2), shape regularity, and $q_F \notin \Gamma_h$ with $H \geq c\ell$. The distinction between $t \in T_{y_F}\Gamma$ and the plane of the projected faces is also $O(\ell)$. By uniform continuity of Π on a compact neighbourhood of $\mathcal{K}$, $\sigma_{\min}(\Pi_{\text{actual}}) \geq s_0/2$ for $\ell \leq h_*$.

Choose x with $t \cdot \Pi x \geq \frac{s_0}{2}|x|$ and normalise. Then

$$t \cdot \sum \mathcal{M} \geq \frac{s_0}{2C_E} |\mathfrak{h}_0| \ell^{5/2} - C\ell^{7/2}.$$

Finally $|\mathfrak{h}_0| = |\mathfrak{h}_{y_F}|$. □

Theorem 4.3 (Curvature-weighted lower bound, $k \geq 3$, Alfeld). *Assume the setting of Thm. B:main on an Alfeld-refined mesh with $k \geq 3$. Let $W = \partial_N u|_\Gamma$ with $\mathfrak{h}W \not\equiv 0$, and let u_h be the velocity of CNS* ($\gamma \geq \gamma_2$) or of GS ($\gamma \geq \gamma_0$). Then for $h_\Gamma \leq h_1$*

$$\|\nabla(\tilde{u} - u_h)\|_{\Omega_h} \geq c h_\Gamma^{3/2} \| |\mathfrak{h}| W \|_{L^2(\Gamma)} - C h_\Gamma^2,$$

with $c = c(\theta, \theta_0, \Lambda_0)$ independent of γ, μ, ρ, h , the pressure and $\varepsilon_h^{(3)}$. This is Thm. B:lb with “ $k \geq 4$ ” replaced by “ $k \geq 3$ ” on Alfeld meshes. It covers saddle points, parabolic points and flat regions ($|\mathfrak{h}| = 0$ there, as the chordal bump vanishes).

Proof. Combine Thm. 4.2 and Lemma 4.1. For $k > 3$ use $Y_3 \subset Y_k$; the moments do not depend on k . □

Remark 4.4 (What the weight means). The norm is the same as in B:lb and cannot be improved to $\|W\|$ in general: on flat parts the sagitta vanishes. Near parabolic points the witness must be taken at the right vertex. A *fixed* vertex choice fails for $\mathfrak{h} = \pm\nu \otimes \nu$ when the opposite edge is $\perp \nu$ (degen.log: “flat square, 1”). This affects semidefinite regions, e.g. cylinders, which the td_k3 remark counted as “definite” but which are not.

5 What is not done

- The constant κ_0 comes from compactness (Thm. 4.2(3)) and is not explicit. A quantitative version of each step of Thm. 3.3 is possible in principle (every division is by ℓ_j , $|F_i|$ or $\mu \cdot s_p$, all of which are bounded below off the degenerate set), but it was not carried out.
- The td_k3 sphere criterion $\tau(\tau + \rho) \leq \frac{1}{2}$ remains the only explicit criterion.
- Worsey–Farin $k = 3$ on general surfaces: not attempted.
- The upper bound keeps the td:open item 3 caveat.

Files (notes/v6/td_k3b; run from this directory; each run < 5 min on 2 cores)

- `ref_nu.py`: exact reference computation for Lemma 1.1; the α and β columns and $\det[\cdot, D]$.
- `face_formula.py` [seed] [trials] $\rightarrow$ `face_formula.log`: per-face exact check against the td_k3 reduced model (29/29).
- `flat_algebra.py` $\rightarrow$ `flat_algebra.log`: symbolic identities of Lemma 2.1.
- `degen.py` $\rightarrow$ `degen.log`: exact, exhaustive-in- $\mathfrak{h}$ degeneracy search.
- `degen_random.py`, `degen_gflat.py` [seed] $\rightarrow$ `degen_random.log`, `degen_gflat.log`: 1783 + 1251 random stars tested against the theorem (run with seeds 5, 7 and 11, 13 for about 4.5 min each).
- `fe_cross.py` $\rightarrow$ `fe_cross.log`: full-FE mod- p cross-check (patchlib).

The scripts import `../td_k3/reduced.py` and `patchlib.py` unchanged.